

GOURISHANKAR SHIRAGUR

Data Engineer | Azure Databricks · PySpark · ADF · Delta Lake · Spark Structured Streaming

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PROFESSIONAL SUMMARY

Data Engineer with 3+ years building production-grade batch and streaming pipelines on Azure Databricks, PySpark, ADF, and Delta Lake. Delivered Bronze-Silver-Gold Lakehouse pipelines processing daily datasets from 10-50 customer sources at HCL Technologies, alongside three independently built pipeline projects that push that experience further into incremental batch design, micro-batch idempotency, and real-time streaming with SCD Type 2. Continued sharpening Databricks and Delta Lake expertise through this applied project work between roles, resulting in a deployed portfolio that mirrors production-grade engineering standards.

Key Strengths: Production Lakehouse pipelines at multi-customer scale | Delta Lake MERGE / SCD Type 2 upsert design | Spark performance tuning (partition pruning, broadcast joins, Z-Ordering) | Manufacturing background enables faster onboarding onto factory/plant data (Industry 4.0) projects

TECHNICAL SKILLS

Big Data & Processing: Apache Spark, PySpark, Spark SQL, Spark Structured Streaming, Delta Lake

Cloud & Platforms: Azure Databricks, Azure Data Factory (ADF), ADLS Gen2 (working knowledge: AWS S3, Glue, EMR)

Data Engineering: Bronze-Silver-Gold / Medallion Architecture, Incremental Processing, CDC, SCD Type 2, Micro-Batch Processing, Metadata-driven Pipelines

Databases & Querying: SQL, Delta Tables, Window Functions, Query Optimization, MERGE/Upserts

Streaming: Structured Streaming, Checkpointing, Watermarking, Exactly-once Processing

Performance Tuning: Partitioning, Broadcast Joins, Caching, Partition Pruning, Z-Ordering, File Compaction

Governance & Quality: Data Lineage, Schema Evolution, Audit Frameworks, Data Quality Validation (working knowledge: Unity Catalog, RBAC)

DevOps: Git, GitHub (working knowledge: Jenkins, CI/CD)

Languages: Python, PySpark, SQL, Java

File Formats: Parquet, JSON, CSV, Delta Tables

PROFESSIONAL EXPERIENCE

Senior Developer, HCL Technologies

Oct 2021 – Mar 2025 | Bengaluru, India

Data Engineering — Azure Databricks / PySpark / ADF / Delta Lake

- Owned Bronze-Silver-Gold batch pipelines on Azure Databricks and ADF, transforming daily snapshot ZIP datasets from 10-50 customer machine-telemetry sources (GPS, fuel, payload, fault codes) into analytics-ready curated outputs.
- Engineered PySpark Silver-layer logic — cleansing, standardization, deduplication, null-handling — with schema validation and business-rule enforcement gating every downstream release across large volumes of daily telemetry records.
- Designed comparison-based incremental-load logic that isolated new and changed records across daily runs, meaningfully cutting reprocessing volume versus full reloads and reducing daily pipeline runtime from hours to well under an hour.
- Drove root-cause analysis and controlled replay/reprocessing of Databricks and ADF job failures within a defined recovery SLA, backed by audit logging and metadata traceability built for fast recovery.
- Established data quality gates — duplicate detection, record-count reconciliation, mandatory-field validation — as a mandatory checkpoint before every Gold-layer publish, keeping data-quality incidents consistently low.
- Served as the primary data-engineering point of contact for analytics teams across 10-50 customer accounts, resolving production data issues and keeping enterprise-scale batch workloads stable.

INDEPENDENT DATA ENGINEERING PROJECTS (2025 – 2026)

Self-directed, deployed projects built to extend HCL production experience into architectures not covered day-to-day — incremental batch design, micro-batch idempotency, and real-time streaming with SCD Type 2.

Incremental Batch Data Pipeline (Bronze–Silver–Gold)

github.com/Gourishshiragur/increment_batch_pipeline

- Designed a production-style incremental batch pipeline on Azure Databricks and Unity Catalog (managed tables and volumes), with an Azure Data Factory orchestration template as a reference integration pattern, modeling daily snapshot fleet-telemetry data at up to ~2 million records per run; validated end-to-end on Databricks Free Edition.
- Implemented Delta Lake MERGE-based upsert logic with hash-based change classification (NEW/CHANGED/UNCHANGED on business fields), achieving a measured, reproducible 93% reduction in reprocessing volume versus full-table reloads, validated end-to-end against control, audit, and reconciliation tables.
- Built Bronze ingestion with schema enforcement and audit logging, Silver-layer cleansing/deduplication, and Gold-layer curated datasets for reporting.
- Backed the pipeline with row-conservation and reconciliation validation across all three layers, a pandas-mirrored unit test suite, and idempotent skip-on-rerun logic (control-table SKIPPED status), deployed via a GitHub Actions CI/CD pipeline with a verified CD job that installs the Databricks CLI and imports notebooks to a live Databricks workspace.

Micro-Batch Processing Pipeline

- Built a retry-safe, replay-safe PySpark micro-batch framework with dual-layer idempotency — a Delta batch-tracker ledger for batch-level skip logic and natural-key Delta MERGE for row-level correctness — verified by a 15-test automated suite (15/15 passing), including 100 replay/retry executions producing zero duplicate writes.
- Implemented dependency-ordered execution with retry handling, plus reconciliation and audit logging across a Bronze-Silver-Gold flow; Databricks orchestrator notebooks and ADF deployment templates define the target Azure architecture, not yet run in a live cloud environment.

Real-Time Streaming Pipeline with SCD Type 2

- Built a production-style Spark Structured Streaming pipeline for real-time banking account-state tracking, combining event-time watermarking, checkpoint-based restart recovery, and an idempotent, single-write SCD2 merge design so replayed or retried micro-batches never duplicate history; measured 2.50s p50 / 3.73s p95 end-to-end latency across a 30-run local benchmark.
- Implemented deterministic SCD Type 2 history rebuild — correctly sequencing multiple same-entity changes within a single micro-batch and re-deriving history for late-arriving events — plus a validated CDC event contract and automatic schema-drift quarantine; separately validated on Databricks Free Edition, with a measured ~9s mean per-batch processing time (9.27s in a clean controlled run, zero data-quality rejections).

EARLIER ENGINEERING EXPERIENCE

Methods Engineer, Mahindra Aerospace Pvt. Ltd.

Jul 2018 – Mar 2019

- Supported aerospace manufacturing operations through process planning, method-sheet preparation, and AutoCAD-based engineering documentation.

Project Engineer, Simha Manufacturing Company

May 2016 – Jul 2017

- Worked on process design, AutoCAD drawings, BOM preparation, and material traceability with shop-floor and vendor coordination.

Production Trainee, Southern Machine Industries, Pune

Feb 2015 – Apr 2016

- Gained hands-on exposure to manufacturing operations, production workflows, and quality documentation practices.

EDUCATION

Bachelor of Engineering (B.E.) — Visvesvaraya Technological University (VTU), Belagavi